(FP1) 3. Summation of series

NO. 1

a. By simplifying $\left(x^n - \sqrt{x^{2n}+1}\right)\left(x^n + \sqrt{x^{2n}+1}\right)$, show that $\frac{1}{x^n - \sqrt{x^{2n}+1}} = -x^n - \sqrt{x^{2n}+1}$. [1]

Let $u_n = x^{n+1} + \sqrt{x^{2n+2}+1} + \frac{1}{x^n - \sqrt{x^{2n}+1}}$.

b. Use the method of differences to find $\sum_{n=1}^{N} u_n$ in terms of N and x. [3]

c. Deduce the set of values of x for which the infinite series

$$u_1 + u_2 + u_3 + \ldots$$

is convergent and give the sum to infinity when this exists. [3]

NO. 2

a. Use standard results from the List of Formulae (MF19) to show that

$$\sum_{r=1}^{n}(7r + 1)(7r + 8) = an^3 + bn^2 + cn$$

where a, b and c are constants to be determined. [3]

b. Use the method of differences to find $\sum_{r=1}^{n} \frac{1}{(7r+1)(7r+8)}$ in terms of n. [4]

c. Deduce the value of $\sum_{r=1}^{\infty} \frac{1}{(7r+1)(7r+8)}$. [1]

NO. 3

Let $S_n = 2^2 + 6^2 + 10^2 + \ldots + (4n - 2)^2$.

a. Use standard results from the List of Formulae (MF19) to show that $S_n = \frac{4}{3}n(4n^2 - 1)$. [4]

b. Express $\frac{n}{S_n}$ in partial fractions and find $\sum_{n=1}^{N} \frac{n}{S_n}$ in terms of N. [4]

c. Deduce the value of $\sum_{n=1}^{\infty} \frac{n}{S_n}$. [1]

NO. 4

a. By first expressing $\frac{1}{r^2-1}$ in partial fractions, show that

$$\sum_{r=2}^{n} \frac{1}{r^2-1} = \frac{3}{4} - \frac{an+b}{2n(n+1)},$$

where a and b are integers to be found. [5]

b. Deduce the value of $\sum_{r=2}^{\infty} \frac{1}{r^2-1}$. [1]

c. Find the limit, as $n \to \infty$, of $\sum_{r=n+1}^{2n} \frac{n}{r^2-1}$. [4]

NO. 5

Let $u_n = \dfrac{4\sin\left(n-\frac{1}{2}\right)\sin\frac{1}{2}}{\cos(2n-1)+\cos 1}$.

i. Using the formulae for cos$P$ ± cos$Q$ given in the List of Formulae MF10, show that

$$u_n = \frac{1}{\cos n} - \frac{1}{\cos(n-1)}.$$ [2]

ii. Use the method of differences to find $\sum_{n=1}^{N} u_n$. [2]

iii. Explain why the infinite series $u_1 + u_2 + u_3 + \ldots$ does not converge. [1]

NO. 6

i. Verify that $\dfrac{2r+1}{r(r+1)(r+2)} = \dfrac{1}{2}\left\{\dfrac{(2r+1)(2r+3)}{(r+1)(r+2)} - \dfrac{(2r-1)(2r+1)}{r(r+1)}\right\}$. [2]

ii. Hence that $\displaystyle\sum_{r=1}^{n} \frac{2r+1}{r(r+1)(r+2)} = \frac{1}{2}\left\{\frac{(2n+1)(2n+3)}{(n+1)(n+2)} - \frac{3}{2}\right\}$. [2]

iii. Deduce the value of $\displaystyle\sum_{r=1}^{\infty} \frac{2r+1}{r(r+1)(r+2)}$. [2]

NO. 7

It is given that $S_n = \sum_{r=1}^{n} u_r$, where $u_r = x^{f(r)} - x^{f(r+1)}$ and $x > 0$.

a. Find S_n in terms of n, x and the function f. [2]

b. Given that f(r) = ln r, find the set of values of x for which the infinite series

$$u_1 + u_2 + u_3 + \ldots$$

is convergent and give the sum to infinity when this exists. [3]

c. Given instead that f(r) = $2 \log_x r$ where $x \neq 1$, use standard results from the List of formulae (MF19) to find $\sum_{n=1}^{N} S_n$ in terms of N. Fully factorise your answer. [4]

NO. 8

Let $u_r = e^{rx}(e^{2x} - 2e^x + 1)$.

a. Using the method of differences, or otherwise, find $\sum_{r=1}^{n} u_r$ in terms of n and x. [3]

b. Deduce the set of non-zero values of x for which the infinite series

$$u_1 + u_2 + u_3 + ...$$

is convergent and give the sum to infinity when this exists. [3]

c. Using a standard result from the list of formulae (MF19), find $\sum_{r=1}^{n} \ln u_r$ in terms of n and x. [3]

NO. 9

Let $S_n = \sum_{r=1}^{n} \ln \frac{r(r+2)}{(r+1)^2}$.

a. Using the method of differences, or otherwise, show that $S_n = \ln \frac{n+2}{2(n+1)}$. [4]

Let $S = \sum_{r=1}^{\infty} \ln \frac{r(r+2)}{(r+1)^2}$.

b. Find the least value of n such that $S_n - S < 0.01$. [3]